X GBOOST for Classification

- Stage 1:

$$\log(\text{odds}) \rightarrow \log_e\left(\frac{P}{1-P}\right) \quad P \rightarrow \text{prob of '1'}$$

$$\log_e\left(\frac{3/5}{1-2/5}\right) = 0.405 \text{ same for all}$$

$$\text{Probability} = \frac{e^{\ln(\text{odds})}}{1+e^{\ln(\text{odds})}} = 0.6.$$

cgpa	placed?	pred1 (prob)	res1
5.70	0	0.6	-0.6
6.25	1	0.6	0.4
7.10	0	0.6	-0.8
8.15	1	0.6	0.4
9.60	1	0.6	0.4

res1 = placed − pred1.

- Stage 2:

cgpa + res1.

cgpa	res1	pred2 (log)	pred2 (prob)	res2.
5.70	-0.6	0.072	0.518	-0.51
6.25	0.4	0.072	0.518	0.48
7.20	-0.8	0.072	0.518	-0.51
8.15	0.4	0.903	0.712	0.28
9.60	0.4	0.903	0.712	0.28.

splitting criteria: 5.97, 6.07, 7.62, 8.87

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$$\text{Similarity Score (SS)} = \frac{(\sum \text{residual}_i)^2}{\sum_{\text{(i=n)}} (\text{prev\_prob}_i (1 - \text{prev\_prob}_i)) + \lambda}$$

i = observation.
no. of total rows.

for now
$\lambda = 0$.

$$SS = \frac{(-0.6+0.4-0.6+0.4+0.4)^2}{5(0.6(1-0.6))+0}$$

Splitting criteria.

1

$\boxed{cgpa < 5.97}$

$\boxed{-0.6}$ $\boxed{0.4,-0.6,0.4,0.4}$

$$SS_L = \frac{(-0.6)^2}{0.6(1-0.6)} = 1.5$$

$$SS_R = \frac{(0.4-0.6+0.4+0.4)^2}{4\times[0.6(1-0.6)]} = 0.37$$

gain = $SS_L + SS_R - SS_{root}$
$= 1.5 + 0.37 - 0$
$= 1.87$.

2

$\boxed{cgpa < 6.07}$

$\boxed{-0.6, 0.4}$ $\boxed{-0.6, 0.4, 0.4}$

gain = 0.13

4

$\boxed{cgpa < 8.87}$

do cal?

3

$\boxed{cgpa < 7.62}$

$\boxed{-0.6, 0.4, -0.6}$ $\boxed{0.4, 0.4}$

gain = 2.22

Higher will be 3

gain = 2.22.

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choose ③.

$$\text{Output} = \frac{\Sigma \text{ residual}_i}{\Sigma(\text{previous\_prob}_i(1 - \text{Prev\_prob}_i)) + \lambda}$$

cgpa < 7.62.

-0.6, 0.4, -0.6 | 0.4, 0.4.

$= 1.66$

$$\text{output} = \frac{-0.6 + 0.4 - 0.6}{3 \times 0.6 \times 0.4}$$

$$= \frac{-0.8}{3 \times 0.24} = -1.11$$

$$\text{pred } m_2 = m1 + \overset{0.3}{\eta} \times m_2.$$

$$\text{pred log} = 0.405 + 0.3 \times \left.\begin{matrix}(-1.11) \\ \text{or} \\ 1.66.\end{matrix}\right\} \text{cgpa} < 7.62.$$

$$\text{pred prob} = \frac{e^{\log_e(\text{odds})}}{1 + e^{\log_e(\text{odds})}}$$

buid model 3 by cgpa + res2.

$m1 + 0.3 \times m_2 + 0.3 \times m_3$] → pred log → pred prob → res3 → models

do this multiple time. to reach 0 or 1